

SUM OF CUBES AND SQUARES

JEE Advanced Preparation — Mathematics

Complete derivations of S_2 and S_3 | Variations for even/odd numbers | Applications | Method of Induction

Subject	Mathematics
Level	JEE Advanced
Topic	Sum of Cubes and Squares — Comprehensive Study
Key Results	$S_2 = n(n+1)(2n+1)/6$; $S_3 = [n(n+1)/2]^2$

1.	Full derivation of $S_2 = n(n+1)(2n+1)/6$ — Method of Differences
2.	Alternate proof of S_2 — Induction
3.	Full derivation of $S_3 = [n(n+1)/2]^2$
4.	Nicomachus theorem — visual proof sketch
5.	Variations: even/odd numbers, partial ranges
6.	Applications in combinatorics (counting squares on board)
7.	6 Fully Solved Worked Examples
8.	10 JEE Advanced Level MCQs
9.	Answer Key with solutions
10.	Key Formulas Quick Reference

1. DERIVATION OF $S_2 = n(n+1)(2n+1)/6$

1.1 Method of Differences (Telescoping Identity)

Step 1: Start with the cubic identity:

$$r^3 - (r-1)^3 = 3r^2 - 3r + 1$$

Proof of identity: $(r-1)^3 = r^3 - 3r^2 + 3r - 1$, so $r^3 - (r-1)^3 = 3r^2 - 3r + 1$.

Step 2: Write the identity for $r = 1, 2, 3, \dots, n$ and add:

$$1^3 - 0^3 = 3(1)^2 - 3(1) + 1$$

$$2^3 - 1^3 = 3(2)^2 - 3(2) + 1$$

$$3^3 - 2^3 = 3(3)^2 - 3(3) + 1$$

... ..

$$n^3 - (n-1)^3 = 3n^2 - 3n + 1$$

Step 3: LHS telescopes (most terms cancel):

$$\text{Sum of LHS} = n^3 - 0^3 = n^3$$

Step 4: Sum of RHS = $3S_2 - 3S_1 + n$

Step 5: Equate and solve:

$$n^3 = 3S_2 - 3n(n+1)/2 + n$$

$$3S_2 = n^3 - n + 3n(n+1)/2 = n(n^2-1) + 3n(n+1)/2$$

$$3S_2 = n(n-1)(n+1) + 3n(n+1)/2 = n(n+1)[(n-1) + 3/2]$$

$$3S_2 = n(n+1)(2n-2+3)/2 = n(n+1)(2n+1)/2$$

RESULT: $S_2 = n(n+1)(2n+1)/6$

1.2 Verification by Mathematical Induction

Base case $n=1$: $S_2 = 1^2 = 1$. Formula: $1*2*3/6 = 1$. [True]

Inductive step: Assume true for $n=k$: $\text{Sum}(r=1 \text{ to } k) r^2 = k(k+1)(2k+1)/6$.

Prove for $n=k+1$:

$$\text{Sum}(r=1 \text{ to } k+1) r^2 = k(k+1)(2k+1)/6 + (k+1)^2$$

$$= (k+1)[k(2k+1)/6 + (k+1)]$$

$$= (k+1)[k(2k+1) + 6(k+1)] / 6$$

$$= (k+1)[2k^2+k+6k+6] / 6$$

$$= (k+1)[2k^2+7k+6] / 6$$

$$= (k+1)(2k+3)(k+2) / 6 = (k+1)(k+2)(2k+3)/6$$

This is exactly the formula with n replaced by $k+1$: $(k+1)(k+2)(2(k+1)+1)/6$. [Proved!]

2. DERIVATION OF $S_3 = [n(n+1)/2]^2$

Method 1: Method of Differences (using 4th power):

Use identity: $r^4 - (r-1)^4 = 4r^3 - 6r^2 + 4r - 1$

Sum from $r=1$ to n (LHS telescopes to n^4):

$$n^4 = 4S_3 - 6S_2 + 4S_1 - n$$

$$4S_3 = n^4 + 6 \cdot n(n+1)(2n+1)/6 - 4 \cdot n(n+1)/2 + n$$

$$4S_3 = n^4 + n(n+1)(2n+1) - 2n(n+1) + n$$

$$4S_3 = n[n^3 + (n+1)(2n+1) - 2(n+1) + 1]$$

$$4S_3 = n[n^3 + (n+1)(2n+1-2) + 1]$$

$$4S_3 = n[n^3 + (n+1)(2n-1) + 1]$$

$$4S_3 = n[n^3 + 2n^2 - n + 2n - 1 + 1] = n[n^3 + 2n^2 + n]$$

$$4S_3 = n^2[n^2 + 2n + 1] = n^2(n+1)^2$$

$$\text{RESULT: } S_3 = n^2(n+1)^2/4 = [n(n+1)/2]^2$$

Method 2: Nicomachus Theorem — The Beautiful Identity

Statement: $1^3 + 2^3 + \dots + n^3 = (1 + 2 + \dots + n)^2$

Visual sketch: Arrange n^3 unit cubes. For $k=1,2,\dots,n$:

$k^3 = k$ copies of k rows of k — can be rearranged into an L-shape of

width k at the bottom-right corner of a $T_n \times T_n$ square (where $T_n = n(n+1)/2$).

All L-shapes together exactly fill the $T_n \times T_n$ square.

Algebraic check ($n=4$): $1+8+27+64 = 100 = 10^2 = [4 \cdot 5/2]^2$ [confirmed!]

This identity is also called the 'Square of Triangular Numbers' theorem.

KEY IDENTITY:

$$S_3 = (S_1)^2 \text{ i.e., } [\text{Sum } r^3] = [\text{Sum } r]^2$$

This is one of the most beautiful identities in combinatorics.

Consequence: S_3 is always a perfect square!

3. VARIATIONS FOR EVEN AND ODD NUMBERS**3.1 Squares of Even Numbers**

$$2^2 + 4^2 + 6^2 + \dots + (2n)^2 = \text{Sum}(r=1 \text{ to } n)(2r)^2 = 4 \cdot \text{Sum}(r^2) = 4S_2$$

$$= 4 \cdot n(n+1)(2n+1)/6 = 2n(n+1)(2n+1)/3$$

Example: $2^2 + 4^2 + 6^2 + \dots + 10^2$ ($n=5$) = $2 \cdot 5 \cdot 6 \cdot 11/3 = 220$

3.2 Squares of Odd Numbers

$$\begin{aligned}1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 &= \text{Sum}(r=1 \text{ to } n)(2r-1)^2 \\&= \text{Sum}(4r^2 - 4r + 1) = 4S_2 - 4S_1 + n \\&= 2n(n+1)(2n+1)/3 - 2n(n+1) + n = n[(2(n+1)(2n+1) - 6(n+1) + 3)] / 3 \\&= n(4n^2 - 1)/3 \\&= \mathbf{n(2n-1)(2n+1)/3}\end{aligned}$$

3.3 Cubes of Even Numbers

$$\begin{aligned}2^3 + 4^3 + 6^3 + \dots + (2n)^3 &= \text{Sum}(2r)^3 = 8 * \text{Sum}(r^3) = 8 * S_3 \\&= \mathbf{8 * [n(n+1)/2]^2 = 2n^2(n+1)^2}\end{aligned}$$

3.4 Cubes of Odd Numbers

$$\begin{aligned}1^3 + 3^3 + 5^3 + \dots + (2n-1)^3 &= \text{Sum}(\text{all cubes } 1 \text{ to } 2n-1) - \text{Sum}(\text{even cubes}) \\&= S_3(2n-1) - 8 * S_3(n-1) \\&= [(2n-1)(2n)/2]^2 - 8 * [(n-1)n/2]^2 \\&= [n(2n-1)]^2 - 2n^2(n-1)^2 \\&= n^2[(2n-1)^2 - 2(n-1)^2] \\&= n^2[4n^2 - 4n + 1 - 2n^2 + 4n - 2] \\&= n^2[2n^2 - 1] \\&= \mathbf{n^2(2n^2 - 1)}\end{aligned}$$

3.5 Partial Range: Sum r^2 from $n+1$ to $2n$

$$\begin{aligned}\text{Sum}(r=n+1 \text{ to } 2n) r^2 &= S_2(2n) - S_2(n) \\&= 2n(2n+1)(4n+1)/6 - n(n+1)(2n+1)/6 \\&= n(2n+1)/6 * [2(4n+1) - (n+1)] \\&= n(2n+1)/6 * [8n+2-n-1] = n(2n+1)(7n+1)/6 \\&= \mathbf{n(2n+1)(7n+1)/6}\end{aligned}$$

4. APPLICATION: COUNTING SQUARES ON A CHESSBOARD

On an $n \times n$ chessboard, the number of squares of size $k \times k$ is $(n-k+1)^2$.
Total squares of all sizes $1 \times 1, 2 \times 2, \dots, n \times n$:
 $= \text{Sum}(k=1 \text{ to } n) (n-k+1)^2 = \text{Sum}(j=1 \text{ to } n) j^2$ [substituting $j=n-k+1$]
 $= S_2(n) = n(n+1)(2n+1)/6$

For standard 8x8 board: $8 \cdot 9 \cdot 17/6 = 204$ squares total

Breakdown: sixty-four 1x1 + forty-nine 2x2 + ... + one 8x8 = $64+49+36+25+16+9+4+1 = 204$

5. WORKED EXAMPLES

Example 1: Find $2^2 + 4^2 + 6^2 + \dots + 30^2$

Solution:

This is the sum of squares of even numbers up to 30 (i.e., $2n=30$, $n=15$)

Formula: $2n(n+1)(2n+1)/3$ with $n=15$

$$= 2 \cdot 15 \cdot 16 \cdot 31 / 3$$

$$= 14880 / 3$$

ANSWER: 4960

Example 2: Find $1^3 + 3^3 + 5^3 + \dots + (2n-1)^3$ (cubes of odd numbers)

Solution:

Formula derived: $n^2(2n^2-1)$

Verification for $n=3$: $1^3+3^3+5^3 = 1+27+125 = 153$

Formula: $9 \cdot (2 \cdot 9 - 1) = 9 \cdot 17 = 153$ [confirmed!]

ANSWER: $n^2(2n^2 - 1)$

Example 3: Prove by induction: $\text{Sum}(r=1 \text{ to } n) r^3 = [n(n+1)/2]^2$

Solution:

Base case $n=1$: $1^3 = 1 = [1 \cdot 2 / 2]^2 = 1$. True.

Inductive step: Assume $\text{Sum}(r=1 \text{ to } k) r^3 = [k(k+1)/2]^2$.

$$\text{Sum}(r=1 \text{ to } k+1) r^3 = [k(k+1)/2]^2 + (k+1)^3$$

$$= (k+1)^2 \cdot [k^2/4 + (k+1)]$$

$$= (k+1)^2 \cdot [(k^2 + 4k + 4)/4]$$

$$= (k+1)^2 \cdot (k+2)^2 / 4$$

$$= [(k+1)(k+2)/2]^2$$

This is the formula for $n=k+1$. PROVED by induction.

Example 4: Find $\text{Sum}(r=1 \text{ to } n)(2r-1)^2$ and verify for $n=4$

Solution:

$$(2r-1)^2 = 4r^2 - 4r + 1$$

$$\text{Sum} = 4 \cdot S_2 - 4 \cdot S_1 + n = 4 \cdot n(n+1)(2n+1)/6 - 4 \cdot n(n+1)/2 + n$$

$$= 2n(n+1)(2n+1)/3 - 2n(n+1) + n$$

$$= n[2(n+1)(2n+1)/3 - 2(n+1) + 1]$$

$$= n(2n-1)(2n+1)/3$$

Verify $n=4$: $1^2+3^2+5^2+7^2 = 1+9+25+49 = 84$

Formula: $4 \cdot 7 \cdot 9 / 3 = 252 / 3 = 84$ [confirmed!]

Example 5: Find the value of $\text{Sum}(r=1 \text{ to } 10)(r^3 - r^2)$

Solution:

$$\text{Sum}(r^3 - r^2) = S_3(10) - S_2(10)$$

$$S_3(10) = [10 \cdot 11 / 2]^2 = 55^2 = 3025$$

$$S_2(10) = 10 \cdot 11 \cdot 21 / 6 = 385$$

$$= 3025 - 385$$

ANSWER: 2640

Example 6: If $S_n = 1^2 + 2^2 + \dots + n^2$, find $S_{20} - S_{10}$

Solution:

$$S_n = n(n+1)(2n+1)/6$$

$$S_{20} = 20 \cdot 21 \cdot 41 / 6 = 17220 / 6 = 2870$$

$$S_{10} = 10 \cdot 11 \cdot 21 / 6 = 2310 / 6 = 385$$

$$S_{20} - S_{10} = 2870 - 385$$

ANSWER: 2485

Note: This equals $\text{Sum}(r=11 \text{ to } 20) r^2 = n(2n+1)(7n+1)/6$ at $n=10 = 10 \cdot 21 \cdot 71 / 6 = 14910 / 6 = 2485$ [confirmed!]

6. PRACTICE QUESTIONS — JEE ADVANCED LEVEL

Q1. [Single Correct] $2^2+4^2+6^2+\dots+20^2$ equals:

- (A) 1540
- (B) 2870
- (C) 1980
- (D) 2380

Q2. [Single Correct] $\sum_{r=1}^n r^3 - [\sum_{r=1}^n r]^2$ is always:

[i.e., $S_3 - (S_1)^2$]

- (A) Positive
- (B) Zero
- (C) Negative
- (D) Depends on n

Q3. [Single Correct] Number of squares of all sizes on a 6×6 board:

- (A) 91
- (B) 100
- (C) 84
- (D) 77

Q4. [Single Correct] $1^3+3^3+5^3$ (first 3 odd cubes) using formula $n^2(2n^2-1)$:

- (A) 120
- (B) 153
- (C) 100
- (D) 145

Q5. [Single Correct] $\sum_{r=1}^{10} (r^3-r^2)$:

- (A) 2480
- (B) 2640
- (C) 2800
- (D) 2560

Q6. [Single Correct] $1^2+3^2+5^2+\dots+19^2$ (sum of first 10 odd number squares):

- (A) 1330
- (B) 1240
- (C) 1540
- (D) 1045

Q7. [Multiple Correct] If $S_n = \sum_{r=1}^n r^2$, then:

- (A) $S_{10} = 385$
- (B) $S_{20} - S_{10} = 2485$
- (C) $S_3 = (S_1)^2$ for all n
- (D) S_n is always divisible by n

Q8. [Single Correct] Sum of cubes of first n even numbers equals:

- (A) $n^2(n+1)^2/2$
- (B) $2n^2(n+1)^2$
- (C) $4n^2(n+1)^2$
- (D) $n(n+1)^2$

Q9. [Single Correct] $\sum_{r=n+1}^{2n} r^2$ simplifies to:

- (A) $n(2n+1)(7n+1)/6$
- (B) $n(n+1)(2n+1)/6$
- (C) $2n(n+1)(2n+1)/3$
- (D) $n^2(n+1)$

Q10. [Single Correct] The value of $\text{Sum}(r=1 \text{ to } 5) r^3$ is:

- (A) 225
- (B) 256
- (C) 169
- (D) 289

7. ANSWER KEY WITH SOLUTIONS

Q#	Answer	Key Solution / Formula Used
Q1	(A) 1540	$2n(n+1)(2n+1)/3$ at $n=10 = 2*10*11*21/3 = 4620/3 = 1540$
Q2	(B) Zero	$S_3 = (S_1)^2$ always — this is Nicomachus's theorem
Q3	(A) 91	$n(n+1)(2n+1)/6$ at $n=6 = 6*7*13/6 = 91$
Q4	(B) 153	$n^2(2n^2-1)$ at $n=3 = 9*(18-1) = 9*17 = 153$
Q5	(B) 2640	$S_3(10)-S_2(10) = 3025-385 = 2640$
Q6	(A) 1330	$n(2n-1)(2n+1)/3$ at $n=10 = 10*19*21/3 = 3990/3 = 1330$
Q7	A, B	(C) is wrong (n^2 vs cube); (D) not always
Q8	(B) $2n^2(n+1)^2$	$8*S_3(n) = 8*[n(n+1)/2]^2 = 2n^2(n+1)^2$
Q9	(A)	$S_2(2n)-S_2(n) = n(2n+1)(7n+1)/6$
Q10	(A) 225	$[5*6/2]^2 = 15^2 = 225$

8. KEY FORMULAS — QUICK REFERENCE

Formula	Closed Form
$S_2 = \text{Sum } r^2 \text{ (r=1 to n)}$	$n(n+1)(2n+1)/6$
$S_3 = \text{Sum } r^3 \text{ (r=1 to n)}$	$[n(n+1)/2]^2 = (S_1)^2$
$\text{Sum } (2r)^2 \text{ (r=1 to n)}$	$2n(n+1)(2n+1)/3$
$\text{Sum } (2r-1)^2 \text{ (r=1 to n)}$	$n(2n-1)(2n+1)/3$
$\text{Sum } (2r)^3 \text{ (r=1 to n)}$	$2n^2(n+1)^2$
$\text{Sum } (2r-1)^3 \text{ (r=1 to n)}$	$n^2(2n^2-1)$
$\text{Sum } r^2 \text{ (r=n+1 to 2n)}$	$n(2n+1)(7n+1)/6$
Squares on $n \times n$ board	$n(n+1)(2n+1)/6$
Nicomachus identity	$\text{Sum } r^3 = [\text{Sum } r]^2$

ADVANCED NOTES: Sum of Cubes and Squares

Deeper Insights on S2 and S3

S2 satisfies: $6 \cdot S_2 = n(n+1)(2n+1) = (2n+1) \cdot n(n+1) = (2n+1) \cdot 2 \cdot T_n$ where T_n is nth triangular.

$S_3 = T_n^2$ where $T_n = n(n+1)/2$ is the nth triangular number.

Interesting: $S_1 + S_2 + S_3 + \dots$ can be summed using generating functions.

JEE 2020: If $a_n = \sum_{k=1}^n (k^2 + k + 1) / (k! + (k+1)!)$, find a_{2019} .

Partial Sums and Telescoping

$\sum_{r=1}^n (r^2 - r) = S_2 - S_1 = n(n+1)(2n+1)/6 - n(n+1)/2 = n(n+1)(2n-2)/6 = n(n-1)(n+1)/3$.

$\sum_{r=1}^n (r^3 - r) = S_3 - S_1 = [n(n+1)/2]^2 - n(n+1)/2 = n(n+1)/2 \cdot [n(n+1)/2 - 1]$.

$= n(n+1)(n^2 + n - 2)/4 = n(n+1)(n+2)(n-1)/4$.

$\sum r(r-1) = \sum (r^2 - r) = n(n-1)(n+1)/3$ — also follows from Vn method.

JEE ADVANCED EXAM STRATEGY

Recognise the Type First: Before computing, classify whether it is AP/GP/AGP/Special series.

Use Standard Results: Memorise S_1 , S_2 , S_3 , and the product series formulas — saves huge time.

Telescoping Check: If T_r has product of consecutive integers, try partial fractions first.

AM-GM Shortcut: For min/max with a constraint, always try AM-GM before calculus.

Verification: Always verify with $n=1$ or $n=2$ before writing final answer.

AGP Infinite Sum: If $|r| < 1$, use $S_{\infty} = a/(1-r) + dr/(1-r)^2$ directly.

Nicomachus: Any time S_3 appears, immediately replace with $(S_1)^2$.

Multiple Correct MCQs: All four options may be true — verify each independently.

COMPLETE FORMULA CONSOLIDATION

#	Series / Expression	Closed Form
1	$\sum r \quad (r=1 \text{ to } n)$	$n(n+1)/2$
2	$\sum r^2 \quad (r=1 \text{ to } n)$	$n(n+1)(2n+1)/6$
3	$\sum r^3 \quad (r=1 \text{ to } n)$	$[n(n+1)/2]^2$
4	$\sum r(r+1)$	$n(n+1)(n+2)/3$
5	$\sum r(r+1)(r+2)$	$n(n+1)(n+2)(n+3)/4$
6	$\sum r \cdot r! \quad (r=1 \text{ to } n)$	$(n+1)! - 1$
7	Sum odd numbers (n terms)	n^2

8	Sum $(2r-1)^2$ ($r=1$ to n)	$n(2n-1)(2n+1)/3$
9	Sum $(2r)^2$ ($r=1$ to n)	$2n(n+1)(2n+1)/3$
10	Sum $(2r-1)^3$ ($r=1$ to n)	$n^2(2n^2-1)$
11	$1/(r(r+1))$ — telescoping	Sum to $n = n/(n+1)$
12	$r/(r+1)!$ — telescoping	Sum to $n = 1-1/(n+1)!$
13	AP n th term	$a+(n-1)d$
14	AP sum n terms	$n[2a+(n-1)d]/2$
15	GP n th term	$a*r^{(n-1)}$
16	GP sum infinity	$a/(1-r)$ for $ r <1$
17	AGP sum infinity	$a/(1-r) + dr/(1-r)^2$
18	$1+2x+3x^2+\dots$ ($ x <1$)	$1/(1-x)^2$
19	Sum $r/2^r$ ($r=1$ to ∞)	2
20	AM $\geq$ GM $\geq$ HM	Always; equality iff all equal